

# CAP. 3

## MODEL EVALUATION AND SELECTION

References:

→ from textbook: cap. 5 - resampling methods

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Text and examples are taken from:

G. James et al., *An Introduction to Statistical Learning: with Applications in R*,  
Springer Texts in Statistics, DOI 10.1007/978-1-4614-7138-7 5,  
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## Chapter 3 - content

- 3.1. Cross-Validation
- 3.2. Bias, variance, and model complexity
- 3.3. The *Bootstrap*

Obs.: topics are numbered as they appear in the reference book.

## Introduction

- **Resampling methods** are indispensable tool in modern statistics.
- **repeatedly drawing samples** from a training set and refitting a model of interest on each sample in order to obtain additional information about the fitted model.
- **Example**, in order to estimate the variability of a linear regression fit:
  - repeatedly draw different samples from the **training data**
  - fit a linear regression to each **new sample**
  - examine the extent to which the resulting fits differ.
- This approach allow one to obtain information that would not be available from fitting the model only once using the original training sample.

- two of the most commonly used resampling methods
  - *cross-validation*
  - *Bootstrap*
- **cross-validation** can be used to estimate the test error associated with a given statistical learning method in order to evaluate its performance, or to select the appropriate level of flexibility.
  - The process of evaluating a model's performance is known as **model assessment**,
  - The process of selecting the proper level of flexibility for a model is known as **model selection**.
- **Bootstrap** is used in several contexts, most commonly to provide a measure of accuracy of a parameter estimate or of a given statistical learning method.

## 5.1 Cross-Validation

### *test error rate:*

- average error that results from using a statistical learning method to predict the response on a new observation.
- Given a data set, we seek for a lower test error.
- test error rate is usually larger than the training error rate.

### *training error rate:*

- can be easily calculated by applying the statistical learning method to the observations used in its training.
- training error rate often can dramatically underestimate the test error rate.

- **The best:** to have a large **designated test set** that can be used to directly estimate the test error rate,
- Otherwise:
  - techniques can be used to estimate this quantity using the **available training data**.
- Now we consider a class of methods that estimate the test error rate by **holding out** a subset of the training observations from the fitting process, and then applying the statistical learning method to those held out observations.



The test set

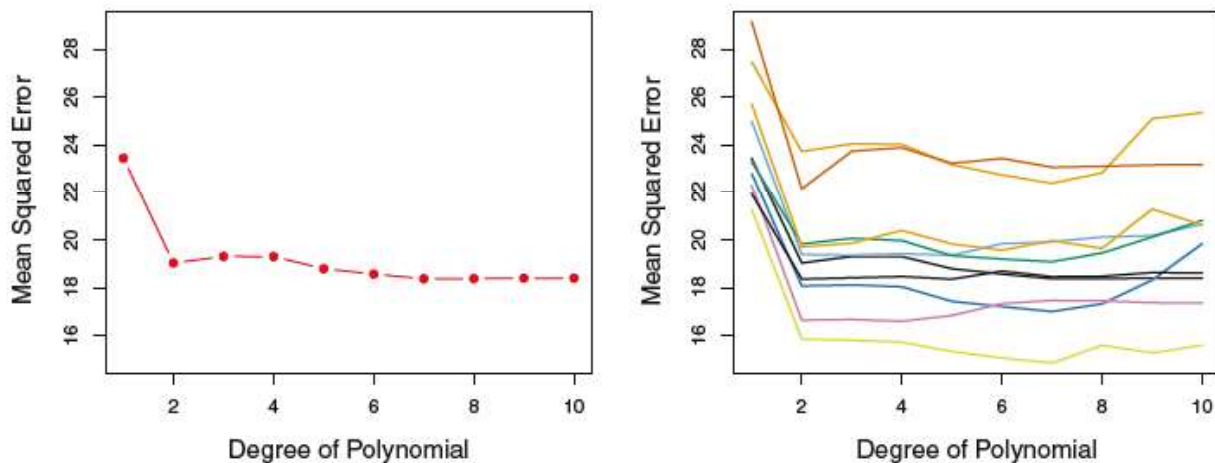
### 5.1.1 The Validation Set Approach



**FIGURE 5.1.** A schematic display of the validation set approach. A set of  $n$  observations are randomly split into a training set (shown in blue, containing observations 7, 22, and 13, among others) and a validation set (shown in beige, and containing observation 91, among others). The statistical learning method is fit on the training set, and its performance is evaluated on the validation set.

- validation set error rate assessed using MSE in the case of a quantitative response → estimate the test error rate.

- **To think about:**
- illustrate the validation set approach on the **Auto** data set
- Try different polinomials
- randomly split the 392 observations into two sets:
  - a training set (196 obs)
  - a validation set (remaining 196 obs.)
- What happens if we repeat the process of randomly splitting the sample set into two parts?
- see Figure 5.2.



**FIGURE 5.2.** The validation set approach was used on the **Auto** data set in order to estimate the test error that results from predicting **mpg** using polynomial functions of **horsepower**. Left: Validation error estimates for a single split into training and validation data sets. Right: The validation method was repeated ten times, each time using a different random split of the observations into a training set and a validation set. This illustrates the variability in the estimated test MSE that results from this approach.

## two potential drawbacks:

- the validation estimate of the test error rate can be highly variable, depending on precisely which observations are included in the training set and which observations are included in the validation set.
- In the validation approach, only a subset of the observations—those that are included in the training set rather than in the validation set—are used to fit the model. Since statistical methods tend to perform worse when trained on fewer observations, this suggests that the validation set error rate may tend to overestimate the test error rate for the model fit on the entire data set.

## 5.1.2 Leave-One-Out Cross-Validation (LOOCV)



**FIGURE 5.3.** A schematic display of LOOCV. A set of  $n$  data points is repeatedly split into a training set (shown in blue) containing all but one observation, and a validation set that contains only that observation (shown in beige). The test error is then estimated by averaging the  $n$  resulting MSE's. The first training set contains all but observation 1, the second training set contains all but observation 2, and so forth.

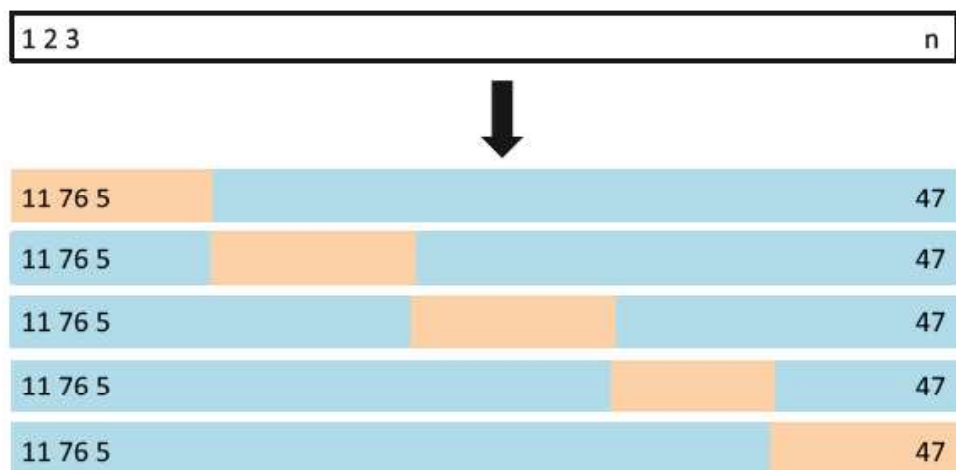
$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n MSE_i$$

### major advantages over the validation set approach:

- it has far less bias.
  - In LOOCV, training sets contain  $n - 1$  observations, almost as many as are in the entire data set.
  - in contrast, in the validation set approach the training set is typically around half the size of the original data set.
  - Consequently, the LOOCV approach tends not to overestimate the test error rate as much as the validation set approach does.
- performing LOOCV multiple times will always yield the same results

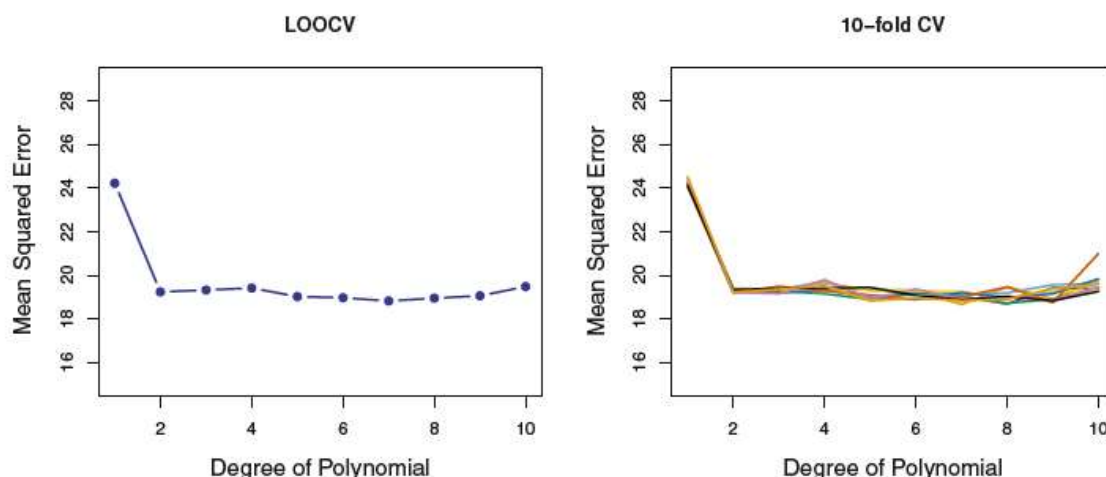
LOOCV is a very general method → can be used with any kind of predictive modeling (ex.: logistic regr.; LDA; etc)

### 5.1.3 *k*-Fold Cross-Validation

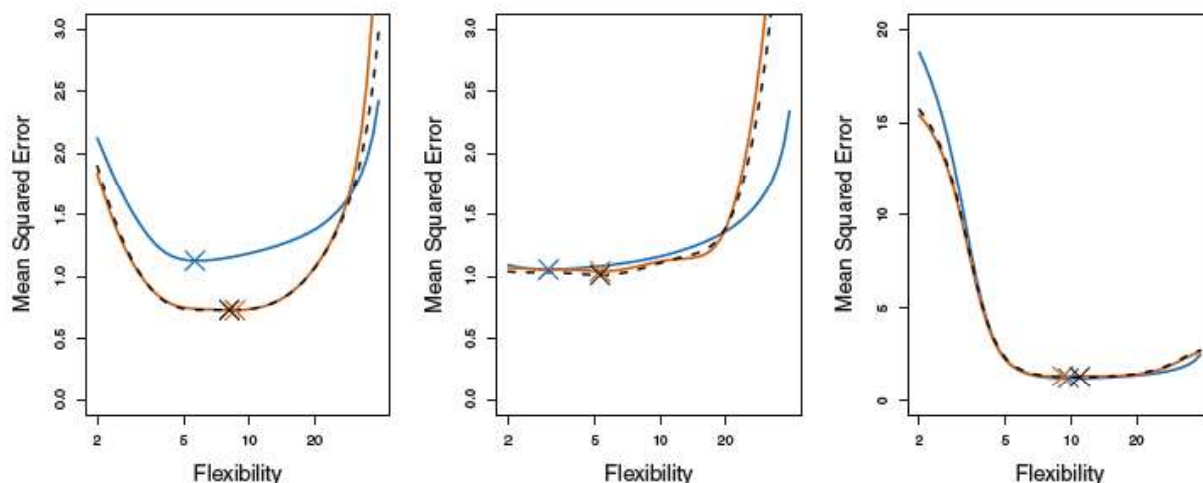


**FIGURE 5.5.** A schematic display of 5-fold CV. A set of  $n$  observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k \text{MSE}_i$$



**FIGURE 5.4.** Cross-validation was used on the **Auto** data set in order to estimate the test error that results from predicting **mpg** using polynomial functions of **horsepower**. Left: The LOOCV error curve. Right: 10-fold CV was run nine separate times, each with a different random split of the data into ten parts. The figure shows the nine slightly different CV error curves.



**FIGURE 5.6.** True and estimated test MSE for the simulated data sets in Figures 2.9 (left), 2.10 (center), and 2.11 (right). The true test MSE is shown in blue, the LOOCV estimate is shown as a black dashed line, and the 10-fold CV estimate is shown in orange. The crosses indicate the minimum of each of the MSE curves.

## So what?

- If want to determine how well a given statistical learning procedure can be expected to perform on independent data → the **actual estimate of the test MSE** is of interest.
- If is performing CV on a number of statistical learning methods, or on a single method using different levels of flexibility, in order to identify the method that results in the lowest test error → interest only in the **location of the minimum point in the estimated test MSE curve**.

### 5.1.4 Bias-Variance Trade-Off for $k$ -Fold Cross-Validation

- $k$ -fold CV with  $k < n$  has a computational advantage to LOOCV.
- But putting computational issues aside, a less obvious but potentially more important advantage of  $k$ -fold CV is that it often gives more accurate estimates of the test error rate than does LOOCV. This has to do with a bias-variance trade-off.
- Remember from pag. 33+

$$E \left( y_0 - \hat{f}(x_0) \right)^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon)$$

### Recall from pag. 33+

- What do we mean by the *variance* and *bias* of a statistical learning method?
- **Variance**: the amount by which  $\hat{f}$  would change if we estimated it using a different training data set. Since the training data are used to fit the statistical learning method, different training data sets will result in a different  $\hat{f}$ .
- **Bias**: the error that is introduced by approximating a real-life problem, which may be extremely complicated, by a much simpler model.

- LOOCV will give approximately **unbiased estimates** of the test error, since each training set contains  $n - 1$  observations, which is almost as many as the number of observations in the full data set.
- And performing  $k$ -fold CV for, say,  $k = 5$  or  $k = 10$  will lead to an **intermediate level of bias**, since each training set contains  $(k - 1)n/k$  observations—fewer than in the LOOCV approach, but substantially more than in the validation set approach.
- Therefore, from the perspective of bias reduction, it is clear that LOOCV is to be preferred to  $k$ -fold CV.

- Besides bias we must also consider the procedure's variance:
- LOOCV has **higher variance than** does  $k$ -fold CV with  $k < n$ .
- the test error estimate resulting from LOOCV tends to have higher variance than does the test error estimate resulting from  $k$ -fold CV.
- Typically, given these considerations, one performs  $k$ -fold cross-validation using  $k = 5$  or  $k = 10$ , as these values **have been shown empirically** to yield test error rate estimates that suffer neither from excessively high bias nor from very high variance

## 5.1.5 Cross-Validation on Classification Problems

- when  $Y$  is qualitative
- CV works as before
- But we use the number of misclassified observations to quantify test error
- The LOOCV error rate takes the form

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{Err}_i$$

Obs.: the CV error rate and validation set error rates are defined analogously

$$\text{Err}_i = I(y_i \neq \hat{y}_i)$$

Example:  
fitting various logistic  
regression models on  
the two-dimensional  
classification data

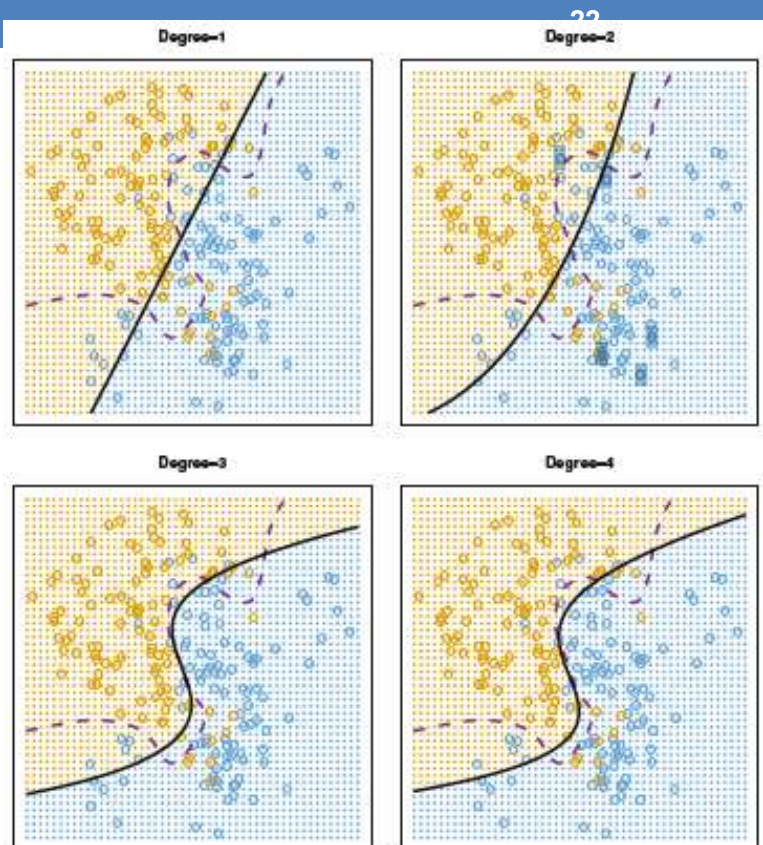
Degree = 1

$$\log \left( \frac{p(X)}{1-p(X)} \right) = \beta_0 + \beta_1 X.$$

Degree = 2

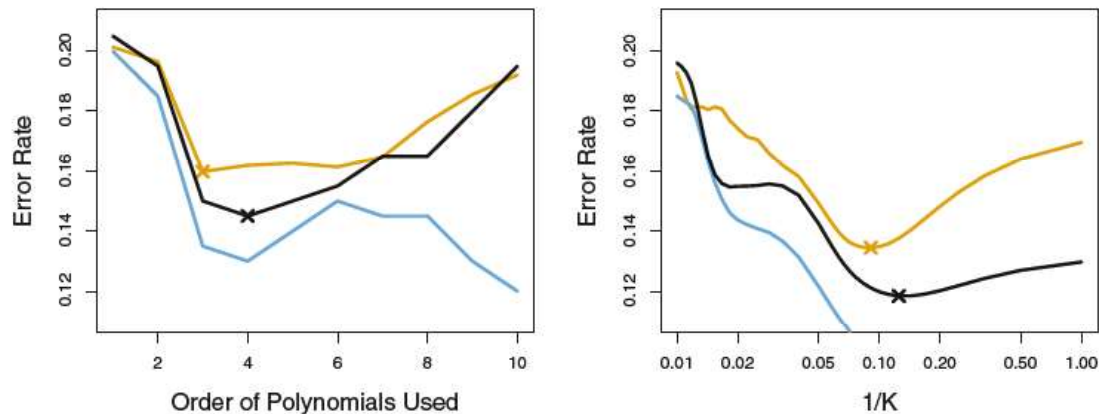
$$\log \left( \frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 X_2^2.$$

Etc...



**FIGURE 5.7.** Logistic regression fits on the two-dimensional classification data displayed in Figure 2.13. The Bayes decision boundary is represented using a purple dashed line. Estimated decision boundaries from linear, quadratic, cubic and quartic (degrees 1–4) logistic regressions are displayed in black. The test error rates for the four logistic regression fits are respectively 0.201, 0.197, 0.160, and 0.162, while the Bayes error rate is 0.133.

- In practice, for real data, the Bayes decision boundary and the test error rates are unknown. So how might we decide between the four logistic regression models?
- → Use cross-validation in order to make this decision.



**FIGURE 5.8.** Test error (brown), training error (blue), and 10-fold CV error (black) on the two-dimensional classification data displayed in Figure 5.7. Left: Logistic regression using polynomial functions of the predictors. The order of the polynomials used is displayed on the x-axis. Right: The KNN classifier with different values of  $K$ , the number of neighbors used in the KNN classifier.

## 5.2 The Bootstrap

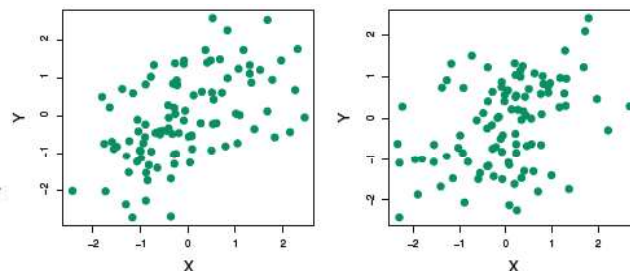
- used to quantify the uncertainty associated with a given estimator or statistical learning method
- Examples:
  - estimate the standard errors of the coefficients from a linear regression fit. (trivial and unnecessary)
  - it can be easily applied to a wide range of statistical learning methods, including some for which a measure of variability is otherwise difficult to obtain and is not automatically output by statistical software
- they illustrate the bootstrap on a toy example in which they wish to determine the best investment allocation under a simple model

- Suppose that we wish to invest a fixed sum of money in two financial assets that yield returns of  $X$  and  $Y$ , respectively, where  $X$  and  $Y$  are random quantities.
- We will invest a fraction  $\alpha$  of our money in  $X$ , and will invest the remaining  $1 - \alpha$  in  $Y$ .
- Since there is variability associated with the returns on these two assets, we wish to choose  $\alpha$  to minimize the total risk, or variance, of our investment. In other words, we want to minimize  $\text{Var}(\alpha X + (1 - \alpha)Y)$ .
- The value that minimizes the risk is given by

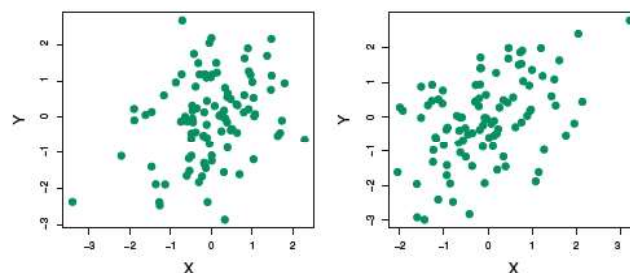
$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

- For a given data set:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$



- Simulations:
- $\sigma_X^2 = 1$ ,
- $\sigma_Y^2 = 1.25$ ,
- $\sigma_{XY} = 0.5$ ,
- so the true value of  $\alpha$  is 0.6



**FIGURE 5.9.** Each panel displays 100 simulated returns for investments  $X$  and  $Y$ . From left to right and top to bottom, the resulting estimates for  $\alpha$  are 0.576, 0.532, 0.657, and 0.651.

- To estimate the standard deviation of  $\hat{\alpha}$  we repeated the process of simulating 100 paired observations of  $X$  and  $Y$ , and estimating  $\alpha$  1,000 times.
- The mean over all 1,000 estimates for  $\alpha$  is

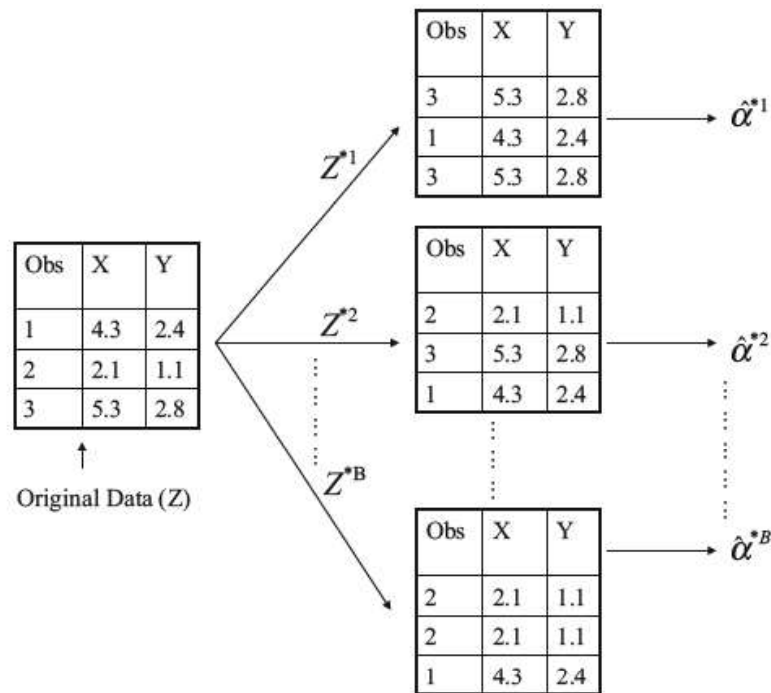
$$\bar{\alpha} = \frac{1}{1,000} \sum_{r=1}^{1,000} \hat{\alpha}_r = 0.5996$$

- the standard deviation of the estimates is

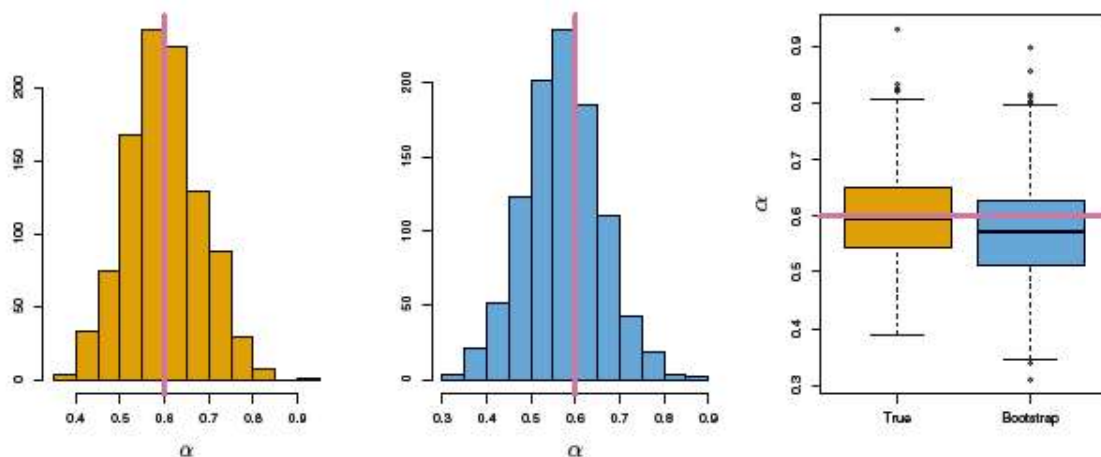
$$\sqrt{\frac{1}{1,000 - 1} \sum_{r=1}^{1,000} (\hat{\alpha}_r - \bar{\alpha})^2} = 0.083$$

- **In practice** → the procedure above cannot be applied, because for real data we cannot generate new samples from the original population.
- Use **bootstrap**: Rather than repeatedly obtaining independent data sets from the population, we instead obtain distinct data sets **by repeatedly sampling observations from the original data set**.
- The sampling is performed **with replacement**.
- For the example the SE of these bootstrap estimates is:

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B - 1} \sum_{r=1}^B \left( \hat{\alpha}^{*r} - \frac{1}{B} \sum_{r'=1}^B \hat{\alpha}^{*r'} \right)^2}$$



**FIGURE 5.11.** A graphical illustration of the bootstrap approach on a small sample containing  $n = 3$  observations. Each bootstrap data set contains  $n$  observations, sampled with replacement from the original data set. Each bootstrap data set is used to obtain an estimate of  $\alpha$ .



**FIGURE 5.10.** Left: A histogram of the estimates of  $\alpha$  obtained by generating 1,000 simulated data sets from the true population. Center: A histogram of the estimates of  $\alpha$  obtained from 1,000 bootstrap samples from a single data set. Right: The estimates of  $\alpha$  displayed in the left and center panels are shown as boxplots. In each panel, the pink line indicates the true value of  $\alpha$ .

## 5.3 Lab: Cross-Validation and the Bootstrap